

Fabrizzio Sabelli

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Education

PhD in Applied Mathematics - Université de Montréal	2025 - Present
Advisor: Lucas Benigni - Centre de Recherches Mathématiques (CRM)	
Master of Science , Applied Mathematics - Université de Montréal	2022 - 2024
Advisor: Maciej Augustyniak - Quantact, Centre de Recherches Mathématiques (CRM)	
Thesis: <i>Numerical Methods for Discrete-time Quadratic Hedging</i> – Mention exceptionnelle	
Bachelor of Science , Mathematics and Computer Science - McGill University	2019 - 2022

Research Items

- **F. Sabelli** (2026) *Homogenization of ℓ_2 -Adversarial Training in High-Dimensions: Exact Dynamics under Stochastic Gradient Descent*, Submitted.
- A. Sokolov, **F. Sabelli**, B. Azadie Faraz, W. Li & A. Seco (2025), *Toward Automating Causal Discovery in Financial Markets and Beyond*, Transactions of ADIA Lab - Interdisciplinary Advances in Data and Computational Science. vol 1:359-392.

Scholarships and Awards

Institut des sciences mathématiques (ISM) Travel Grant (500\$)	2026
Institut des sciences mathématiques (ISM) Travel Grant (500\$)	2025
Université de Montréal Artificial Intelligence Award (10 000\$)	2025
FRQNT Master's Award (26 667\$)	2023 - 2024
NSERC Fin-ML Master's Bursary (7500\$)	2022 - 2023
Université de Montréal Solutions mieux-être LifeWorks Bursary (2500\$)	2022 - 2023

Research Talks & Posters

<i>High-dimensional limit of SGD for ℓ_2-adversarial training of GLMs</i> , Spring school TU Darmstadt, Darmstadt	2026
<i>On The High-Dimensional Limit of ℓ_2-Adversarial Regression</i> , Poster, ML Theory Summer School, Princeton	2025
<i>Numerical Methods for Discrete-Time Quadratic Hedging</i> , CRM Quantact Day seminar, Montréal	2024
<i>Leveraging Large Language Models for ESG Investing in Finance</i> , Fin-ML 5-year milestone seminar, Montréal	2023

Teaching Experience

Teaching Assistant, Université de Montréal, <i>Probability</i>	Fall 2025
Teaching Assistant, Université de Montréal, <i>Stochastic Processes</i>	Fall 2024
Teaching Assistant, Université de Montréal, <i>Stochastic Processes</i>	Summer 2024
Teaching Assistant, Université de Montréal, <i>Stochastic Processes</i>	Fall 2023
Teaching Assistant, Université de Montréal, <i>Probability</i>	Winter 2023
Teaching Assistant, Université de Montréal, <i>Probability and Statistics</i>	Winter 2023
Teaching Assistant, Université de Montréal, <i>Multivariable Calculus</i>	Fall 2022

Research Experience and Internships

Research Internship - RiskLab, University of Toronto	2023 - 2024
Advisor: Luis Seco, Department of Mathematics, University of Toronto	
Subject: Efficient causal graph discovery using large language models with applications in finance	
Research Internship - McGill University, Montréal	2021
Advisor: Jérôme Waldispühl, McGill School Of Computer Science, McGill University	
Subject: Graph neural networks for link prediction of non-canonical base pairings in RNA molecules	

Work Experience

Data Scientist Intern - Sibli, Montréal	January 2023 - August 2023
Software Development Intern - Explorance, Montréal	Summer 2022
Software Development Intern - Explorance, Montréal	Summer 2021

Academic Community Involvement

Student Intersectoral Committee - Institute for Data Valorization (IVADO), Montréal	2023 - Present
Quantact Student Committee - Quantact, Université de Montréal, Montréal	2024

Competitions

Caisse de dépôt et placement du Québec Polyfinances Datathon - 2nd place	November 2023
Montreal Data Science Competition - 1st place	September 2023
National Bank of Canada Polyfinances Datathon - 3rd place	November 2022